Feedback Information-Guided Spectral Variability Attention Network for Hyperspectral Unmixing

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***Abstract*— Hyperspectral images (HSIs) encounter an inherent challenge due to spectral variability (SV), which directly impacts the accuracy of endmember extraction and abundance estimation. Most unmixing networks based on autoencoder (AE) overlook the impact of SV and instead focus more on exploring the features of endmembers. In this article, we propose a feedback information-guided SV attention network (FSVA-Net), an AE-based neural network that utilizes a feedback information-guided structure to exploit and model the SV factors. We develop a feedback enhancement module (FEM) that utilizes residuals from a linearly reconstructed image to reweight and emphasize pixels affected by SV. And an SV attention (SVA) mechanism is proposed to explore interference-affected information in a global perspective according to the enhanced feature. With the extracted SV-related feature, we design a generative augmented linear mixing model (ALMM)-based decoder to model SV from the perspective of scaling and perturbation in a more reliable way. In a more concern on SV architecture, the proposed network achieves excellent performance on both synthetic and real datasets.**

***Index Terms*— Attention mechanism, autoencoder (AE) network, feedback, hyperspectral unmixing, spectral variability (SV).**

NOMENCLATURE

$\mathbf{Y}$	Original hyperspectral data.
$\boldsymbol{E}$	Embedding of original data.
$\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}$	Query, key, and value matrices of $\boldsymbol{E}$.
$\boldsymbol{X}$	Feature map of input data.
$\boldsymbol{Y}_p$	Linearly reconstructed image.
$\boldsymbol{E}_p$	Embedding of linearly reconstructed image.
$\boldsymbol{E}_c$	Complementary feature.
$\boldsymbol{Q}_c$	Query matrix of complementary feature.
$\boldsymbol{X}_{\text{sv}}$	Attention-weighted spectral variability feature.

I. INTRODUCTION

HYPERSPECTRAL image (HSI) is a widely used technique that contain abundant spectral–spatial information [1]. Various researches have been conducted in the field of HSI processing, such as unmixing, image classification, band selection, and change detection [2], [3], [4]. The spatial resolution of HSI is limited due to the limitations of remote sensing imaging hardware. Therefore, a single pixel often contains multiple materials, which is referred to as a mixed pixel [5]. The goal of hyperspectral unmixing is to decompose mixed pixels into their fundamental components and corresponding proportions, called endmembers and abundances, which is often used to analyze constituents in mixed pixels.

To decompose mixed pixels into endmembers and abundances, it is necessary to study the mixing effects between spectra. Based on the reflection effects encountered by photons during their propagation, spectral mixing models can be divided into linear mixing model (LMM) [6] and nonlinear mixing models (NLMMs) [7]. LMM assumes that photons are reflected only once by the ground object before reaching the imaging device, resulting in an obtained spectrum that is a linear combination of endmembers. While the LMM intuitively describes the influence of the spatial distribution of materials on the imaging spectrum, it overlooks the impact of photon interactions on the spectrum, leading to significant errors in applications. NLMMs provide a solution by modeling the nonlinear effects between spectra to mitigate the influence of photon interactions on unmixing performance [8]. Various mixing models considering the nonlinear mixing effect have been developed. The Hapke model [9], [10] was based on the approximation of isotropic multiple scattering to model the nonlinear scattering of photons on planetary surfaces. Additionally, bilinear mixing models [11], [12], [13] and multilinear mixing models [14], [15], [16] also attempt to mitigate the influence of photon interactions on unmixing accuracy by modeling high-order nonlinear mixing effects between endmembers.

LMM offers a physically interpretable model for hyperspectral unmixing, while it lacks the ability to deal with complex conditions in real scenarios. NLMMs capture complex interactions, but their complexity makes parameter estimation challenging. Furthermore, the mentioned models assume that the spectra of endmembers in real scenes are constant, which conflicts with actual conditions. During remote sensing imaging, factors such as illumination and terrain can cause variations in the spectral signatures of the same material [17]. Additionally, inherent factors like plant growth can also lead to changes in material spectra. Hence, unmixing algorithms based on spectral variability (SV) have been developed, which could be divided into four categories: 1) endmember bundles [18], [19], [20], [21]; 2) statistical probability distribution [22], [23], [24], [25]; 3) modified LMMs [26], [27], [28], [29];

Manuscript received 9 July 2024; revised 12 August 2024; accepted 28 August 2024. Date of publication 30 August 2024; date of current version 12 September 2024. The work of Shuhan Chen was supported in part by the National Natural Science Foundation of China under Grant 62171404. *(Corresponding author: Shuhan Chen.)*

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Digital Object Identifier 10.1109/TGRS.2024.3452323

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and 4) autoencoder (AE)-based algorithms [30], [31], [32], [33], [34].

Algorithms based on endmember bundles argue that describing a kind of material with a single spectrum is insufficient due to the influence of SV, which can introduce significant errors. Therefore, representing a specific material with an endmember bundle can mitigate this issue, where each spectrum in the bundle could be the reflectance of an endmember instance. Multiple endmember spectral mixture analysis (MESMA) [35] can select suitable endmembers from a large number of endmember bundles to represent the current pixel and estimate abundance, which offers an effective solution for bundle-based unmixing. Bateson et al. [18] proposed the endmember bundle extraction (EBE) algorithm, which divided the data into multiple subsets, applied an endmember extraction algorithm to each subset, and then integrated the extraction results from each subset into an endmember bundle. Convex geometry theory-based endmember extraction algorithms are often used to construct endmember bundles, while these methods may make it challenging to capture endmembers located at the boundary of the simplex. Recent researches [19], [20], [21] tend to dynamically capture endmember variability in datasets and utilize multiobjective particle swarm optimization to select the optimal combination of endmembers.

Another kind of unmixing method introduces the capability to address SV by imposing probabilistic statistical constraints on endmembers. Normal compositional model (NCM) [22] assumed that endmember variables follow a Gaussian distribution, which extracted endmembers from datasets using vertex component analysis (VCA) [36] as the mean of the distribution. Beta compositional model (BCM) [23] considered endmember variability as variables following a Beta distribution. However, Zhou et al. [24] argued that using Gaussian distribution to represent endmember variability is insufficient and proposed using Gaussian mixture models. Liu et al. [25] constructed composite prior distributions for abundance and endmember variability based on temporal continuity and spectral smoothness, achieving adaptive constraints on various conditions.

Modified LMMs (MLMMs) are based on the LMM and enhance its ability to model SV by introducing different interference terms. From the perspective of regular and irregular disturbances, extended LMM (ELMM) [26] and perturbed LMM (PLMM) [27] introduced scaling and perturbation terms to the LMM, respectively. Building upon ELMM, generalized LMM (GLMM) [29] proposed per-band endmember scaling to account for nonuniform influences. Augmented LMM (ALMM) [28] designs scaling terms in pixel-wise and an SV dictionary to address both regular and irregular interference. Multitype mixing model [37] simultaneously models nonlinear effects and SV in LMM. Spectral–spatial intra-class variability model [38] takes into account the variability from both spectral and spatial perspectives. These models provide effective methods for addressing SV by improving the LMM.

Deep learning (DL) recently has been widely applied in HSI processing due to its powerful data-fitting capability. AE was introduced to hyperspectral unmixing due to its self-supervised characteristic and demonstrated excellent performance [39]. uDAS [40] designed a denoising AE and applied a sparsity constraint to the encoder to achieve AE-based unmixing. DAEN [41] utilized stacked AEs to learn spectral features and provide initialization information for unmixing. EGU-Net [42] designed a dual-stream Siamese network to extract pure pixel features, guiding the training of the unmixing network. Networks that focus solely on pixel-level features are insufficient to meet the performance requirements of unmixing, as they overlook the spatial information in the images [43], [44], [45], [46].

Local homogeneity is a common attribute of images, and the convolution neural network (CNN) is an effective tool for extracting local information. Consequently, CNN is widely used in unmixing networks. CNNAEU [43] designed a CNN-based unmixing network that preserves the spatial structure of the image while extracting features. MiSiCNet [44] and RDAE [45] further enhanced model stability by incorporating residual structures, in addition to using CNN layers to extract spatial information. Due to the 3-D tensor architecture of HSIs, 2D-CNNs may overlook the rich information in the spectral dimension. 3D-CNN-var [32] designed a 3D-CNN-based encoder to capture spatial–spectral information. To constrain unrelated spatial information, GTCAN [47] fused gated features based on the 3D-CNN.

It is important to note that CNNs tend to extract local features, while recent studies have shown that transformers [48], [49], [50], [51] demonstrate the positive contribution of utilizing global features for image processing. DeepTrans [52] leveraged a vision transformer (ViT) block to capture global feature dependencies, demonstrating the potential of transformers in unmixing. DGMSAN [34] designed a transformer-based encoder to map HSIs into latent space and introduced a discriminator to enhance endmember extraction quality. ULA-Net [33] divided the image into patches and designed a transformer concentrating on the central pixel to selectively extract effective information. UST-Net [53] and UnDAT [54] utilized several Swin Transformer blocks to extract deep features, enriching contextual information for feature enhancement.

AE-based models that consider SV typically design specific structures to either mitigate the impact of SV or model SV explicitly. DeepGUn [30] utilized a generative model to learn the low-dimensional representation of endmember variability and better model the variability. PGMSU [31] proposed a dual-branch network, which could simultaneously generate endmember by the variational AE (VAE) [55] and estimate the abundance for every pixel. Many AE-based algorithms try to consider endmember variability by modifying their decoder parts. 3D-CNN-var [32] applied variational inference in the decoder to consider the uncertainty properties in endmembers. ULA-Net [33] designed a decoder based on the ELMM to scale the endmembers for different pixels. MAAE [56] introduced a learnable parameter to mitigate the influence of scaling caused by varying illumination.

A. Motivation

Decoders based on MLMM utilize the powerful representation capabilities of DL and achieve data-driven adaptive SV

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modeling. 3D-CNN/add [57] and MAAENet [56] designed decoders based on PLMM and ELMM, aiming to reduce the influence of endmembers and uncertainty on unmixing. However, PLMM adds additional perturbed endmembers to each endmember, which may lead to ill-posed problems. On the other hand, the ELMM model only introduces scaling terms, limiting its ability to handle irregular perturbations. ALMM effectively addresses these issues. However, its modeling of both spectral scaling and perturbations under partial bands poses challenges for convergence of the training procedure. In this article, we propose a module to take advantage of the strong modeling ability of ALMM and attempt to avoid the adverse effects of SV.

Besides, developing an effective network to extract features related to SV is needed. Most previous works tend to focus on exploit the spatial–spectral features in data, which neglect the specific characteristic of SV. The distribution of SV is random and complex, and existing algorithms lack research on separating SV from original data. We consider the characteristics of material spectra and SV to be distinct, and applying feature extraction algorithms designed for SV can further enhance unmixing performance. The feedback mechanism makes adaptive adjustments to training procedures by implementing posterior knowledge propagation and is widely used in image processing. Therefore, the feedback mechanism offers an approach to design a suitable network structure to balance the modeling of pure spectra and SV terms.

Transformers [48], [49], [50] have provided significant assistance in improving the performance of HSIs processing, thanks to their modeling capability for long-range dependencies. However, the patch-based transformer block in DeepTrans [52] is known for its high computational complexity. In UST-Net [53], the sparse attention mechanism based on shifted windows alleviates this issue. Nevertheless, its window partitioning method is data-agnostic, which can result in the omission of useful information and the misapplication of irrelevant information. Inspired by deformable convolutions [58], we propose an SV attention (SVA) mechanism to adaptively focus on regions of interest and mitigate the interference caused by redundant information on accuracy.

B. Contributions

In response to the above, we proposed the feedback information-guided SVA network (FSVA-Net) to separate and explicitly model SV information in images, aiming to eliminate the impact of SV on unmixing accuracy. The model employs a feedback enhancement mechanism to utilize the reconstruction residuals from linear unmixing to guide SV modeling. Additionally, an SVA mechanism and a generative ALMM-based decoder are proposed to selectively explore SV information, modeling it as scaling and perturbation terms. In conclusion, the contributions of our work can be summarized as follows.

1) In this article, we present the FSVA-Net, which aims to leverage feedback information between the linearly reconstructed image and the original data to guide feature extraction and modeling for SV.
2) Distinguished from directly using residuals, we custom design a feedback enhancement module (FEM) to transform residuals into a mask. This mask serves as an indicator of the challenging reconstruction parts of linear unmixing, specifically the portions affected by SV. It further weights the data embedding to obtain complementary features.
3) We design an SVA mechanism, which can adaptively and selectively achieve global sampling from the deep features of data under the guidance of complementary features. This mechanism selects and extracts information from relevant regions, avoiding information redundancy.
4) We design a generative ALMM-based decoder that models SV considering both regular scaling and irregular perturbations. Different from modeling based on spectral dictionary in ALMM, the decoder generates perturbation terms based on probabilistic generation. This extends the decoder's ability to model SV and reduces its impact on unmixing accuracy.

The rest of this article is organized as follows. Section II provides an introduction to related works, including mixing models considering SV and AE-based unmixing networks. In Section III, we present a detailed description of our proposed FSVA-Net, which comprises an FEM module, an SVA module, and a generative ALMM-based decoder. Section IV discusses experimental results and provides comparisons with state-of-the-art algorithms. Finally, Section V concludes this article.

II. Related Works

The proposed model is constructed based on spectral mixing models and AEs. In this section, we will introduce the related works of our proposed algorithm.

A. Spectral Mixing Models

1) Linear Mixing Model: To explicitly character the mixing effect between the spectra of materials, the LMM assumes that the observed spectra are linear combination of endmembers. Thus, we represent the observed HSI, endmember, abundance, and noise in matrix form as $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \ldots, \mathbf{y}_N] \in \mathbb{R}^{L\times N}$, $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \mathbf{m}_3, \ldots, \mathbf{m}_P] \in \mathbb{R}^{L\times P}$, $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \ldots, \mathbf{a}_N] \in \mathbb{R}^{P\times N}$ and $\mathbf{R}_n \in \mathbb{R}^{L\times N}$. L represents the number of bands, P denotes the number of endmembers, and N indicates the number of pixels in the data. And the mathematical equation of the LMM is described as

$$\begin{aligned} &\mathbf{Y} = \mathbf{MA} + \mathbf{R}_n \\ &\text{s.t. } \mathbf{1}_P^{\mathrm{T}}\mathbf{A} = \mathbf{1}_N^{\mathrm{T}}, \quad \mathbf{A} \geq 0, \ \mathbf{M} \geq 0 \end{aligned} \tag{1}$$

where the dot product represents the linear mixing of endmembers in accordance with the abundance. In addition, the model should meet the sum-to-one constraint and nonnegative constraint to ensure physical realizability.

2) Augmented LMM: ALMM modifies the LMM by introducing scaling term and SV dictionary to model both regular and irregular interference. $\mathbf{R}_s \in \mathbb{R}^{N\times N}$ represents the scaling term for every pixel with its diagonal elements $s \geq \mathbf{0}$.

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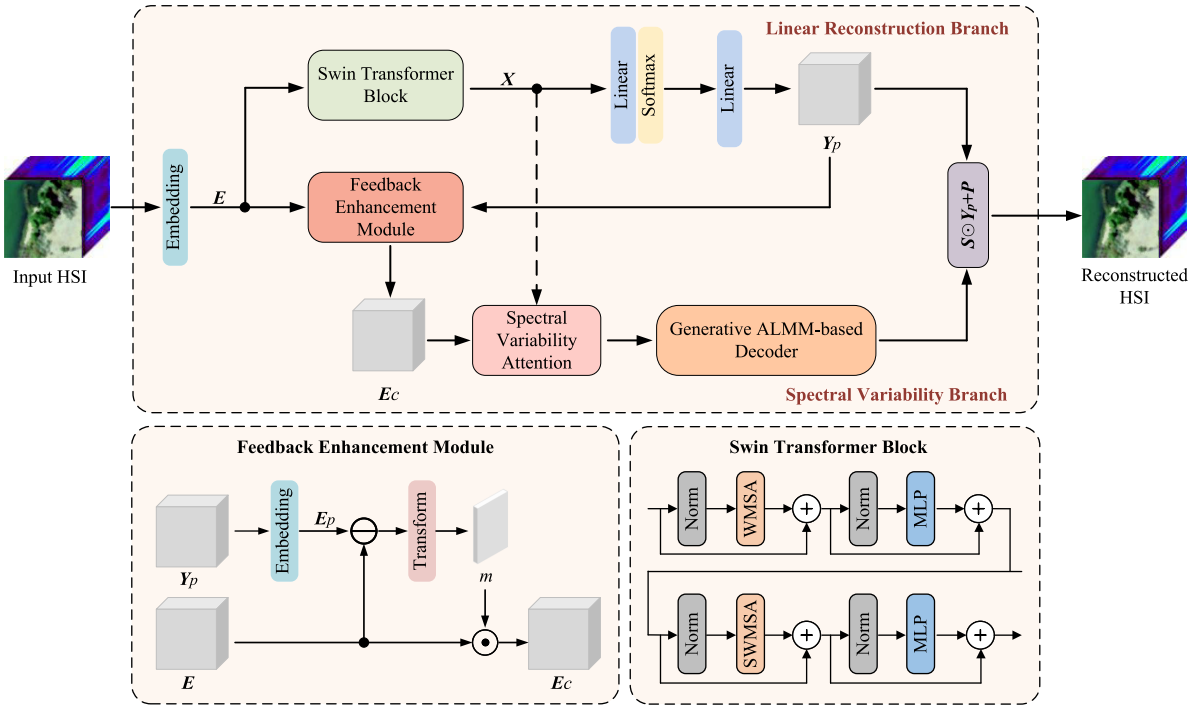


Fig. 1. Architecture of the proposed FSVA-Net that utilizes the feedback structure to construct a linear reconstruction branch and an SV branch. The linear reconstruction branch consists of a Swin Transformer block and linear mixing module. The SV branch attempts to exploit useful information from the discrepancy between data embeddings and linearly reconstructed spectral cube, which consists of an FEM, an SVA module, and generative ALMM-based decoder. The reconstructed HSI cube is obtained according to the mixing model considering scale and perturbation interference.

The expressions $\mathbf{R}_p = [\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3, \ldots, \boldsymbol{r}_{D_p}] \in \mathbb{R}^{L \times D_p}$ and $\mathbf{B} = [\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3, \ldots, \boldsymbol{b}_N] \in \mathbb{R}^{D_p \times N}$ denote SV dictionary and corresponding coefficient matrix. And D_p represents the total number of basis vectors in matrix $\mathbf{E}$. The formula of ALMM is described as

$$\mathbf{Y} = \mathbf{MAR}_s + \mathbf{R}_p\mathbf{B} + \mathbf{R}_n \tag{2}$$

where $\mathbf{S}$ models the spectral scaling effect caused by SV. And the SV dictionary $\mathbf{R}_p$ takes into account irregular spectral signatures, enhancing the model's flexibility in handling variabilities.

B. AE-Based Unmixing Networks

Due to their capability of achieving low-dimensional representations of data and end-to-end training, AEs are commonly employed in hyperspectral unmixing. Specific designs of encoders and decoders enable the network to effectively explore valuable information according to characteristics of HSIs and reconstruct the data based on mixing models.

In AEs, the encoder compresses the input into low-dimensional latent variables, and then the decoder maps the features back into the original data. In the context of hyperspectral unmixing theory, it is commonly considered that the low-dimensional representations stand for the abundance and the weights of the decoder represent the endmembers of input data. We conclude the process of AEs in unmixing as

$$f_{\text{encoder}}(\mathbf{Y} \mid \Theta_e) = \mathbf{A} \tag{3}$$

$$f_{\text{decoder}}(\mathbf{A} \mid \Theta_d) = \widehat{\mathbf{Y}} \tag{4}$$

where $f_{\text{encoder}}(\cdot)$ and $f_{\text{decoder}}(\cdot)$ represent the networks of the encoder and the decoder, which extract features and reconstructs the original data. Explicitly, the output of the encoder is the abundance matrix $\mathbf{A}$. And $\widehat{\mathbf{Y}}$ denotes the reconstructed data. Based on the loss between the input data and the reconstructed one, the network can be trained in a self-supervised manner. Furthermore, AEs have great potential for feature extraction by designing encoder structures according to the characteristics of the data.

III. Proposed Method

This section presents an introduction to our proposed network, which consists of an FEM, an SVA module, and a generative ALMM-based decoder. To enhance readers' comprehension of this article, we provide a detailed explanation of the abbreviations used in Nomenclature section.

A. Network Architecture

The overall architecture of the network is shown in Fig. 1, which can be seen as a dual-branch network consisting of a linear reconstruction branch and an SV branch. In our work, we attempt to separate interference from pure features. Thus, the pure features can be considered as linear combinations of endmembers, and we refer to it as a linearly reconstructed image. Before processing the hyperspectral data $\mathbf{Y} \in \mathbb{R}^{L \times H \times W}$, we implement dimensionality reduction in order to reduce computational complexity. We utilize 1×1 convolutions to linearly map the data to embeddings, which can be

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formulated as

$$\boldsymbol{E} = \text{Norm}(\text{Conv}(\mathbf{Y})) \tag{5}$$

where $\boldsymbol{E} \in \mathbb{R}^{D\times H\times W}$ is the data embedding. And D is the dimension of the data after dimensionality reduction and is set to 32. Additionally, normalization is applied to adjust the data distribution and stabilize the training process of the network. This approach enables us to extract useful spectral band information while preserving the spatial information of the image.

In this article, we consider the features of linearly reconstructed images to be easy to capture, while the accuracy of hyperspectral unmixing is limited due to factors such as SV and noise interference. Previous networks tend to design various modules and exploit material spectral characteristics and high-level feature information in images, neglecting the modeling of randomly distributed and complex mixing SV. Therefore, we attempt to design a simple network branch to extract valuable information about endmembers from the input and model them based on the LMM. Additionally, we feed the difference between linearly reconstructed images and data embeddings $\boldsymbol{E}$ back to the SV branch to model SV.

In the linear reconstruction branch, we employ a Swin Transformer block to exploit useful global information, which is present in Fig. 1 in detail. Unlike classical Transformers, Swin Transformer computes the correlations within and between windows in a shifting window manner, reducing the complexity of the model. In this module, the data is divided into windows in the size of $K \times K$ and used for subsequent attention weight calculations (K is set to 5 here and can be adjusted in practice according to the data). In both window-based multihead self-attention (WMSA) and shifted WMSA (SWMSA) mechanisms, data embeddings are subject to the following processing. For data $\boldsymbol{E}_i \in \mathbb{R}^{K^2\times D}$ in each window, it is linearly projected into query, key, and value matrices $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$

$$\boldsymbol{Q} = \boldsymbol{E}_i \boldsymbol{W}_1^{\text{Q}}, \quad \boldsymbol{K} = \boldsymbol{E}_i \boldsymbol{W}_1^{\text{K}}, \ \boldsymbol{V} = \boldsymbol{E}_i \boldsymbol{W}_1^{\text{V}} \tag{6}$$

where $\boldsymbol{W}_1^{\text{Q}}$, $\boldsymbol{W}_1^{\text{K}}$, and $\boldsymbol{W}_1^{\text{V}} \in \mathbb{R}^{D\times d}$ are projection operators that transform the input embeddings into matrices with channel dimension of d. And $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V} \in \mathbb{R}^{K^2\times d}$, where $d = D/h$ (the default value of h is 4) depends on the number of heads h in multihead self-attention (MSA) mechanism. Then the attention-weighted feature is calculated according to the MSA mechanism as

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\boldsymbol{T}}}{\sqrt{d}}\right)\boldsymbol{V} \tag{7}$$

where $\sqrt{d}$ in the denominator is used to prevent potential gradient vanishing and explosion, thus ensuring more stable network training.

In the Swin Transformer block, to achieve interaction between features at different levels and model complex relationships between features, the model introduces skip connections and a multilayer perception (MLP). The information flow in WMSA can be represented as follows:

$$\boldsymbol{E}_i' = \text{WMSA}(\boldsymbol{E}_i) + \boldsymbol{E}_i \tag{8}$$

$$\boldsymbol{E}_i'' = \text{MLP}(\boldsymbol{E}_i') + \boldsymbol{E}_i'. \tag{9}$$

Unlike WMSA, SWMSA deploys a shifted window according to the step size of $(K/2, K/2)$. Therefore, dependencies can be calculated across different parts of the data, enabling the construction of global relationships. The remaining operations are the same as WMSA and can be described as follows:

$$\boldsymbol{E}_i''' = \text{SWMSA}(\boldsymbol{E}_i'') + \boldsymbol{E}_i'' \tag{10}$$

$$\boldsymbol{X}_i = \text{MLP}(\boldsymbol{E}_i''') + \boldsymbol{E}_i''' \tag{11}$$

where $\boldsymbol{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n] \in \mathbb{R}^{D\times H\times W}$ denotes the feature map of the input, which concatenates all the windows in the Swin Transformer block. And n is the number of windows.

Then, we construct a simple design to achieving unmixing. A 1×1 convolution layer is used here to estimate the abundance and the linearly reconstructed image $\boldsymbol{Y}_p \in \mathbb{R}^{L\times H\times W}$, which is equal to a single linear layer $\boldsymbol{W}_1 \in \mathbb{R}^{P\times D}$. The processing is shown as follows:

$$\boldsymbol{A} = \text{Softmax}(\boldsymbol{W}_1\boldsymbol{X} + \boldsymbol{b}) \tag{12}$$

$$\boldsymbol{Y}_p = \boldsymbol{W}_2\boldsymbol{A} \tag{13}$$

where $\boldsymbol{A} \in \mathbb{R}^{P\times H\times W}$ is the abundance cube, and nonlinear activation function Softmax can map the value of $\boldsymbol{A}$ into $0 \sim 1$. $\boldsymbol{b}$ is the bias vector, and the weight matrix $\boldsymbol{W}_2 \in \mathbb{R}^{L\times P}$ can be seen as the endmember matrix $\boldsymbol{M}$.

In this article, we do not regard the linear combination of endmembers and abundances as the reconstruction of the original image $\boldsymbol{Y}$; instead, we consider it as representing the pure components within it. Obviously, there will be differences between the original data embedding $\boldsymbol{E}$ and linearly reconstructed image $\boldsymbol{Y}_p$. We attribute this to the presence of SV factors. And if $\boldsymbol{Y}_p$ is regarded as the reconstructed image, it will inevitably introduce errors. Thus, we introduce our SV branch, which utilizes an FEM, an SV mechanism, and a generative ALMM-based decoder to capture and model SV. We will present the above modules in the following parts.

B. Feedback Enhancement Module

The feedback mechanism is a process that enables information at high-level be transmitted to earlier stages. It can convey posterior information to the earlier part of the network to adjust and improve the network training process, which is widely adopted in various fields. In our work, we design an FEM to exploit SV terms potentially distributed in the input HSI.

In the FEM, we attempt to capture SV features by utilizing the residual guidance between the $\boldsymbol{Y}_p$ (linear mixing of endmembers) and the original data embedding $\boldsymbol{E}$. To avoid the interference of redundant spectral information on the feedback signal, we employ a feedback enhancement mechanism on the features. We map the linearly reconstructed image $\boldsymbol{Y}_p$ to the feature space in the same way as 5. We formulated the operation as follows:

$$\boldsymbol{E}_p = \text{Norm}(\text{Conv}(\boldsymbol{Y}_p)) \tag{14}$$

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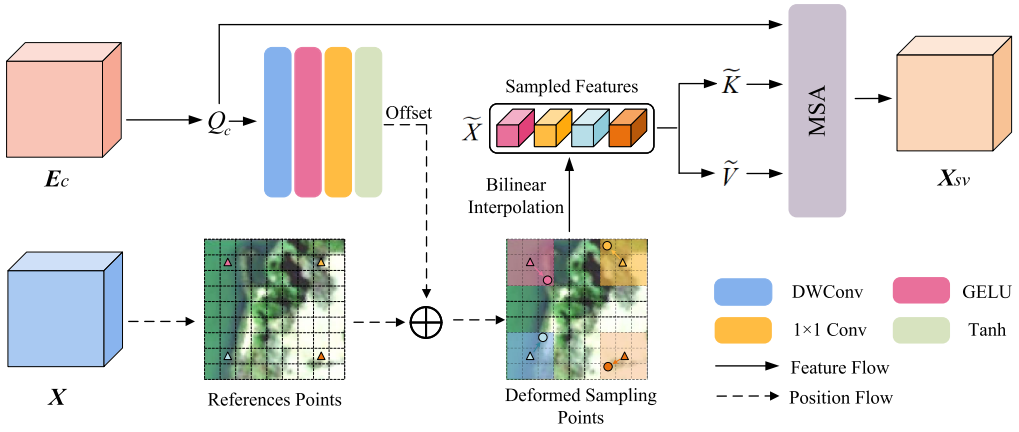


Fig. 2. Network of the proposed SVA mechanism. The complementary feature $\boldsymbol{E}_c$ is utilized as query vectors to guide the selection of sampling points. The module samples from the feature map of the input image, effectively capturing valuable information associated with SV. Four windows from the model are presented as examples.

where $\boldsymbol{E}_p \in \mathbb{R}^{D\times H\times W}$ represents the embedding of the linearly reconstructed image in the channel dimension of D.

Then, we feed $\boldsymbol{E}_p$ back to the earlier stage of the network and calculate the residual between $\boldsymbol{E}$ and $\boldsymbol{E}_p$. We can consider the discrepancy between them as interferences that may be caused by the SV. Therefore, we calculate the root mean square (rms) for each pixel and employ the normalization and the nonlinear function Sigmoid to transform the difference into a mask. In the mask, values are distributed between 0 and 1, representing the degree to which they are affected by SV. The transformation operation can be represented by the following equations:

$$\text{Res} = \text{Norm}(\text{rms}(\boldsymbol{E} - \boldsymbol{E}_p)) \tag{15}$$

$$\text{rms}(\boldsymbol{E} - \boldsymbol{E}_p)_{(i,j)} = \sqrt{\frac{1}{D}\sum_{k=1}^{D}(\boldsymbol{E} - \boldsymbol{E}_p)_{k,i,j}^2} \tag{16}$$

$$\boldsymbol{m} = \text{Sigmoid}(\text{Res}) \tag{17}$$

where rms refers to the calculation of the rms across the channel dimension for each position. Equation 16 represents the expression for calculating the rms along the channel dimension at position (i, j). This process, combined with normalization to prevent baseline shifts, produces the normalized residual measurement matrix Res $\in \mathbb{R}^{H\times W}$, ensuring the model's performance. And $\boldsymbol{m} \in \mathbb{R}^{H\times W}$ is the weighted mask of every position. The bigger value in $\boldsymbol{m}$ indicates that this position may be influenced by SV heavily. On the contrary, the smaller one shows a lower likelihood of the presence of SV in this pixel. Thus, we impose the mask on the feature maps of each channel in the data embedding $\boldsymbol{E}$ and obtain the complementary feature

$$\boldsymbol{E}_c = \boldsymbol{m} \odot \boldsymbol{E} \tag{18}$$

where $\boldsymbol{E}_c \in \mathbb{R}^{D\times H\times W}$ is the complementary feature, and $\odot$ represents the Hadamard product. $\boldsymbol{E}_c$ contains features that complement the endmember characteristics, namely SV information, hence we refer to it as complementary features. Instead of directly utilizing differences to guide the lower branch, we opt for a masking mechanism to enhance relevant regions associated with SV while suppressing irrelevant areas. This way, $\boldsymbol{E}_c$ can be utilized to guide subsequent exploration of SV information.

C. SVA Mechanism

In real datasets, SV is difficult to capture due to its random distribution and complex mixing. Global attention computations offer a solution to address SV. However, dense attention mechanisms like ViT incur high computational costs due to the attention computation over the entire image, and they may be susceptible to interference from irrelevant information. With its sparse attention mechanism, Swin Transformer effectively alleviates the computational complexity by reducing redundancy. The distribution of SV information is more dispersed than that of endmembers. However, Swin Transformer's window partitioning, which is data-independent, may overlook essential information necessary for long-range modeling. Deformable receptive fields present an approach to capturing SV information. This module generates global reference points based on input feature maps and offsets the sampling points using complementary features. By adaptively adjusting the attention area based on the characteristics of the input feature map and complementary feature, the proposed method achieves global attention in a data-dependent way. The architecture of the proposed SVA mechanism is shown in Fig. 2.

In our proposed SVA mechanism, we aim to employ the complementary feature $\boldsymbol{E}_c$ to guide the extraction of the SV feature in the feature map $\boldsymbol{X}$. Referring to the window partitioning in Swin Transformer, we divide the feature map along the spatial dimension into multiple windows. The number of windows $N_w = H_w \times W_w$ is determined by the factor r (we set the factor $r = 3$ here), such that $H_w = H/r$ and $W_w = W/r$. According to the number of windows, the network generates coordinate grid of points $p \in \mathbb{R}^{2\times H_w\times W_w}$ at equal intervals. p is the 2-D coordinate value of sampling points. Based on the size of the windows, the coordinate values in p are normalized to the range of $[-1, +1]$, where $(-1, -1)$ indicates the top-left corner of the window, and $(+1, +1)$ indicates the bottom-right corner of the window.

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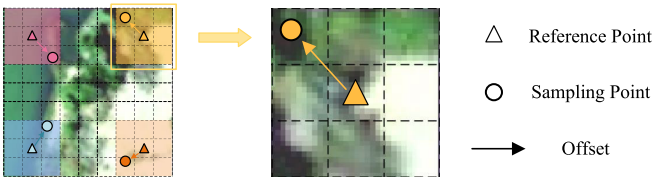


Fig. 3. Illustration of deformed sampling points in SV attention. We show the situation in one of the windows as an example.

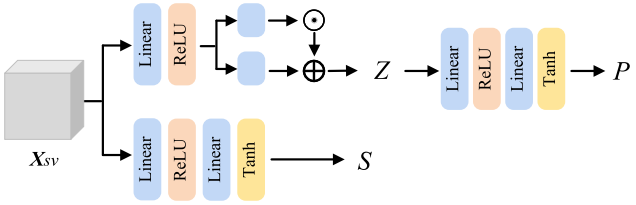


Fig. 4. Architecture of the generative ALMM-based decoder. It considers both scaling and irregular interference to model the potential influence of SV.

Due to the enhancement of the variability information in $\boldsymbol{E}_c$, it can offer more effective information. We choose $\boldsymbol{E}_c$ as the query matrix to compute dependencies with the sampling points in the feature map and extract information related to variability. Therefore, the complementary features $\boldsymbol{E}_c$ are linearly projected as the query matrix $\boldsymbol{Q}_c$. To keep the expression concise, some simple reshape operations are omitted

$$\boldsymbol{Q}_c = \boldsymbol{E}_c \boldsymbol{W}_2^{\mathrm{Q}} \tag{19}$$

where $\boldsymbol{W}_2^{\mathrm{Q}} \in \mathbb{R}^{D\times d}$ is a learnable projection matrix, and $\boldsymbol{Q}_c \in \mathbb{R}^{N\times d}$ is the query matrix of the complementary features. d represents the dimension of vectors in $\boldsymbol{Q}_c$, which is the same in Swin Transformer block.

Additionally, following the concept of data-dependent deformable receptive fields, the lightweight offset generation network $f_{\mathrm{offset}}(\cdot)$ adjusts the positions of the sampling points by generating offsets. The architecture of $f_{\mathrm{offset}}(\cdot)$ is shown in Fig. 2. The network employs depthwise convolutions to capture spatial information while reducing computational complexity. Additionally, 1×1 convolutions are used to modify the channel dimensions of features. Furthermore, the network incorporates the GELU function to enhance its nonlinear modeling capabilities. The offsets $\triangle p$ are obtained by mapping the offsets through the Tanh function to the range of $(-1, +1)$. Furthermore, to enhance the stability of the network, we manually set the maximum offset value as factor f_s

$$\triangle p = f_s \cdot f_{\mathrm{offset}}(\boldsymbol{Q}_c) \tag{20}$$

where $\triangle p \in \mathbb{R}^{2\times H_w \times W_w}$ is the offset cube, with its values representing the magnitude of the offsets for corresponding sampling points.

Based on the generated offset values, we obtain the positions of the sampled points after shifting. Since the magnitude of the offsets is likely to be decimal, we employ bilinear interpolation to determine the values of the sampled points

$$\widetilde{\boldsymbol{X}} = \phi(\boldsymbol{X}, p + \triangle p) \tag{21}$$

$$\phi(\boldsymbol{X}, (p_x, p_y)) = \sum_{(r_x, r_y)} g(p_x, r_x) g(p_y, r_y) \boldsymbol{X}[:, r_y, r_x] \tag{22}$$

where $\phi(\cdot, \cdot)$ is the bilinear interpolation function and $g(a, b) = \max(0, 1 - |a - b|)$. (r_x, r_y) represents the location in $\boldsymbol{X}$ and (p_x, p_y) indexes the sampled point. $\widetilde{\boldsymbol{X}}$ denotes the sampled features. To visualize the process of deformable sampling in this module, we draw the schematic in Fig. 3. In the diagram, triangles represent reference sampling points, whose position is one of the values in p. Arrows indicate the offset values from $\triangle p$ for the reference sampling points. Circles represent the final sampling points after applying the offsets. And then, we can conduct an MSA mechanism on them to capture variability information, which can be described as

$$\widetilde{\boldsymbol{K}} = \widetilde{\boldsymbol{X}} \boldsymbol{W}_2^{\mathrm{K}}, \quad \widetilde{\boldsymbol{V}} = \widetilde{\boldsymbol{X}} \boldsymbol{W}_2^{\mathrm{V}} \tag{23}$$

$$\boldsymbol{X}_{\mathrm{sv}} = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_c \widetilde{\boldsymbol{K}}^{\boldsymbol{T}}}{\sqrt{d}}\right) \widetilde{\boldsymbol{V}} \tag{24}$$

where $\widetilde{\boldsymbol{K}}, \widetilde{\boldsymbol{V}} \in \mathbb{R}^{N_w \times d}$ are the key and value matrices of the sampled features, and N_w is the number of windows in SVA. $\boldsymbol{X}_{\mathrm{sv}} \in \mathbb{R}^{D\times H\times W}$ is an attention-weighted SV feature. Thus, keys and values are selected in a data-dependent way. The network can dynamically focus on areas related to variability under the guidance of the complementary feature, enabling SV attention.

D. Generative ALMM-Based Decoder

To deal with variability in HSIs, we design a generative ALMM-based decoder, which is shown in Fig. 4 and consists of a perturbation branch and a scaling branch. As for the scaling branch in the lower part of Fig. 4, we denote it as f_{scaling}, which utilizes a multilayer fully connected network with nonlinear transformation to extract scaling information from SV-enhanced feature $\boldsymbol{X}_{\mathrm{sv}}$. We describe the progress as follows:

$$\boldsymbol{S} = f_{\mathrm{scaling}}(\boldsymbol{X}_{\mathrm{sv}}) \tag{25}$$

where $\boldsymbol{S} \in \mathbb{R}^{H\times W}$ is the scaling factor indicating the degree of scaling for each pixel.

As for the perturbation branch, ALMM employs a high-dimensional variability dictionary to model the perturbation terms. However, due to the complexity and diversity of SV, as well as the limited representation capacity of the variability dictionary, the unmixing performance of ALMM is constrained. We attempt to model additive variability factors from a probabilistic distribution perspective, but their distribution is complex and difficult to estimate. Thus, we incorporate variational inference techniques, such as the VAE, which is a generative model and can enhance the diversity of the perturbation terms generated by the model.

Specifically, we aim to map the features to a multivariate normal distribution to capture more complex latent space distributions and achieve a more diverse representation. This is achieved by generating mean and variance vectors and weighting them onto a random vector sampled from a standard

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normal distribution, which can be denoted as

$$\boldsymbol{\mu}, \boldsymbol{\sigma} = f_{\phi}(\boldsymbol{X}_{\mathrm{sv}})$$
$$\boldsymbol{Z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \tag{26}$$

where $\boldsymbol{\mu}, \boldsymbol{\sigma} \in \mathbb{R}^{N \times d_l}$ are the generated mean matrix and variance matrix of the latent space, $N = H \times W$ is the number of pixels, and d_l is the dimension of the latent vector. $\boldsymbol{\epsilon}$ is a vector sampled from a standard normal distribution, and $\boldsymbol{Z} \in \mathbb{R}^{N \times d_l}$ is the latent variable.

Thus, diverse samples are encoded into a multivariate normal distribution. Then, we utilize an inference network $f_{\text{perturbation}}$ to decode the representation into perturbation terms, which can be described as

$$\boldsymbol{P} = f_{\text{perturbation}}(\boldsymbol{Z}) \tag{27}$$

where $\boldsymbol{P} \in \mathbb{R}^{L \times H \times W}$ is the estimated perturbation term, which contains additive SV influence of each pixel. We introduce scaling terms and perturbation terms to the LMM, and the network outputs reconstructed images

$$\widehat{\boldsymbol{Y}} = \boldsymbol{S} \odot \boldsymbol{Y}_p + \boldsymbol{P} \tag{28}$$

where $\widehat{\boldsymbol{Y}} \in \mathbb{R}^{L \times H \times W}$ is the reconstructed image. And pixel-wise scaling factors are applied to the linearly reconstructed image $\boldsymbol{Y}_p$.

E. Loss Function

The design of the loss function is crucial for the performance of networks. An appropriate loss function can effectively guide the network toward convergence. To measure the similarity between reconstructed images and original images in both numerical and structural aspects, we utilize mean square error (mse) and spectral angle distance (SAD) as the reconstruction error

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \left(\widehat{\mathbf{Y}}_i - \mathbf{Y}_i \right)^2 \tag{29}$$

$$\mathcal{L}_{\mathrm{SAD}} = \frac{1}{N} \sum_{i=1}^{N} \arccos \left(\frac{\left\langle \mathbf{Y}_i, \widehat{\mathbf{Y}}_i \right\rangle}{\|\mathbf{Y}_i\|_2 \left\| \widehat{\mathbf{Y}}_i \right\|_2} \right) \tag{30}$$

$$\mathcal{L}_{\mathrm{re}} = \alpha \mathcal{L}_{\mathrm{MSE}} + \beta \mathcal{L}_{\mathrm{SAD}}. \tag{31}$$

Due to the dispersed distribution of materials in real scenarios, abundance maps are often sparse. Moreover, as we integrate scaling and perturbation terms to model SV, we assume that perturbation terms are primarily distributed within specific spectral bands. Therefore, the L_1-norm is introduced to impose sparse constraints on both the abundance and perturbation terms

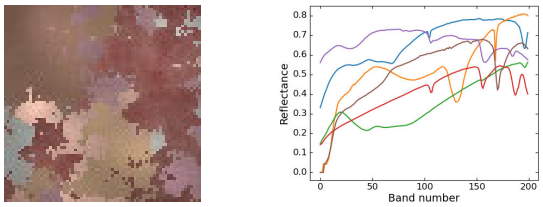
$$\mathcal{L}_{\mathrm{sparsity}} = \gamma \|\mathbf{a}\|_1 + \delta \|\mathbf{p}\|_1$$
$$= \gamma \sum_{i=1}^{P} |a_i| + \delta \sum_{j=1}^{L} |p_j|. \tag{32}$$

To better control the latent space, enrich the distribution of generated samples, and enhance the generative capacity of the model, KL divergence is employed to impose constraints on the latent variables. By minimizing the KL divergence, the variational distribution can approach the desired prior distribution. This facilitates learning effective latent representations and enables modeling and generative capabilities for the data [31], [55]

$$\mathcal{L}_{\mathrm{kl}} = \lambda \mathcal{D}_{\mathrm{KL}}(q_{\phi}(z_i | x_{\mathrm{sv},i}) || p(z))$$
$$= -\frac{\lambda}{2} \sum_{i=1}^{d_l} \left[\log \boldsymbol{\sigma}_z^{(i,i)} - \boldsymbol{\mu}_z^{(i)2} - \boldsymbol{\sigma}_z^{(i,i)2} + 1 \right] \tag{33}$$

where $q_{\phi}(z_i | x_{\mathrm{sv},i})$ is the variational distribution, and $p(z)$ is the prior distribution of latent variable. The overall loss function can be summarized as

$$\mathcal{L}_{\mathrm{all}} = \mathcal{L}_{\mathrm{re}} + \mathcal{L}_{\mathrm{sparsity}} + \mathcal{L}_{\mathrm{kl}}. \tag{34}$$



(a) (b)

Fig. 5. Synthetic dataset. (a) Pseudo-color image. (b) Endmembers.

IV. Experiments

In this section, experiments are conducted on both a synthetic dataset and real datasets. We compare our proposed model with several state-of-the-art algorithms to present its superiority. Additionally, an ablation study is conducted to demonstrate the effectiveness of our proposed model.

A. Hyperspectral Dataset

1) Synthetic Dataset: The synthetic dataset is sourced from the United States Geological Survey (USGS) mineral spectral library. To accurately represent the material distribution, we generate abundance data using a Gaussian field-based toolbox [59]. And scaling factors that follow a 2-D Gaussian distribution are utilized to scale pixels. Furthermore, Gaussian noise is added to simulate noise present in the scenes. The data format is $70 \times 70 \times 200$, and the pseudo-color image and endmember curves are shown in Fig. 5(a) and (b).

2) Samson Dataset: Fig. 6(a) displays the pseudo-colored images derived from the Samson dataset [60]. The dataset consists of a grid of size 95×95 pixels, with each pixel containing 156 channels. The images represent three endmembers of materials: soil, trees, and water.

3) Jasper Ridge Dataset: Fig. 6(b) illustrates the Jasper Ridge dataset [60]. The dataset comprises a grid of size 100×100 pixels, with each pixel containing 198 bands. The dataset encompasses four distinct endmembers: tree, water, soil, and road.

4) Urban Dataset: Fig. 6(c) presents the Urban dataset, which consists of 307×307 pixels, with each pixel containing 162 spectral bands. This dataset includes six different endmembers: asphalt, grass, tree, roof, metal, and dirt.

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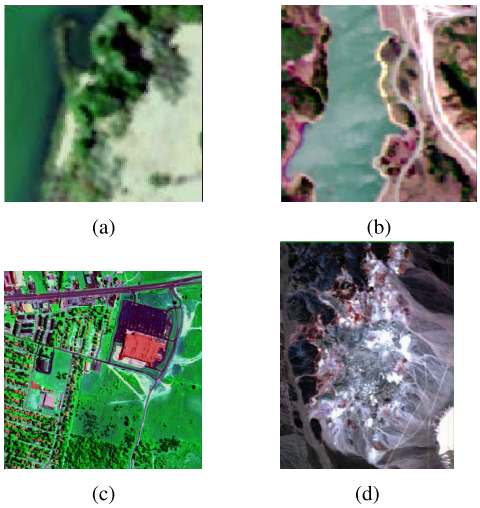


Fig. 6. Real dataset. (a) Samson dataset. (b) Jasper Ridge dataset. (c) Urban dataset. (d) Cuprite dataset.

5) Cuprite Dataset: The Cuprite dataset, shown in Fig. 6(d), consists mainly of various minerals and includes a grid of size 250 × 190 pixels, with each pixel containing 188 channels. This image consists of 12 endmembers and only provides ground truth for the endmember spectra.

B. Baselines and Metrics

1) Baselines: We provide a brief introduction to eight comparative algorithms used in experiments. The experimental results are obtained by repeating the algorithms ten times under the parameters recommended by the authors and then averaging the results. The comparative algorithms are as follows.

1) *FCLS [61]:* Linear unmixing considering abundance sum-to-one and nonnegative constraints.
2) *PLMM [27]:* Modified LMM with consideration of perturbation terms to deal with SV.
3) *ELMM [26]:* An approach employing a scaling term to model SV.
4) *ALMM [28]:* An augmented LMM with scaling term and SV to overcome both scaling and irregular interference.
5) *PGMSU [31]:* A generative network generates endmembers for each pixel while estimating abundance.
6) *DeepTrans [52]:* An AE-based network utilizing self-attention mechanism to exploit long-range relationships.
7) *UST-Net [53]:* A U-shaped transformer network to capture the global material distribution and distant interdependencies.
8) *ULA-Net [33]:* An AE-based network introducing spatial–spectral features double fusion mechanism and designing a scaling decoder.

2) Metrics: When evaluating the performance of hyperspectral unmixing algorithms, various aspects can be taken into account. These include the accuracy of abundance estimation and the precision of endmember extraction achieved by the model. In this article, we use the root mean square error of abundances (aRMSE) to evaluate its performance

$$\text{aRMSE} = \sqrt{\frac{1}{NP}\sum_{i=1}^{N}\sum_{j=1}^{P}\left(a_{ji}-\widehat{a}_{ji}\right)^2} \tag{35}$$

where the true abundance of the jth endmember in the ith pixel is denoted as a_{ji}, while the estimated abundance is represented as $\widehat{a}_{ji}$. The variables N and P correspond to the total number of pixels and the number of endmembers, respectively. The performance of endmember extraction is evaluated using the endmember SAD (eSAD)

$$\text{eSAD} = \frac{1}{P}\sum_{j=1}^{P}\arccos\left(\frac{\mathbf{m}_j^T\widehat{\mathbf{m}}_j}{\|\mathbf{m}_j\|_2\|\widehat{\mathbf{m}}_j\|_2}\right) \tag{36}$$

where $\mathbf{m}_j$ denotes the true spectrum of the jth endmember, while $\widehat{\mathbf{m}}_j$ represents the estimated spectrum. To comprehensively evaluate the performance of endmember extraction, we introduce the spectral information divergence of the endmembers (eSID) as an evaluation metric

$$\text{eSID} = \frac{1}{P}\sum_{i=1}^{P}\left(\sum_{j=1}^{L} m_{i,j}\log\frac{m_{i,j}}{\widehat{m}_{i,j}} + \sum_{k=1}^{L}\widehat{m}_{i,k}\log\frac{\widehat{m}_{i,k}}{m_{i,k}}\right) \tag{37}$$

where $m_{i,j}$ and $m_{i,k}$ represent the values of the jth and kth bands in the ith true spectrum, respectively. The representation of the estimated spectra $\widehat{m}_{i,j}$ and $\widehat{m}_{i,k}$ follows the same convention. And L represents the number of bands in the current image.

C. Hyperparameter Settings

Hyperparameter settings can adjust the adaptability of the network to different datasets. The proposed network considers reconstruction, sparsity, and variational constraints. To ensure a balance between numerical and structural emphasis during model reconstruction, we set hyperparameters α and β to control the emphasis on MSE and SAD, respectively. γ and δ are utilized to assign weights to the sparsity constraints applied to abundances and perturbation terms, respectively. Additionally, λ works as the weight of variational constraint.

To ensure the network achieves optimal performance, we employed a grid search method within a reasonable range to find suitable parameters. Furthermore, during the parameter optimization process, we can employ a hierarchical optimization strategy. In the loss function, the reconstruction loss plays a decisive role. Therefore, we first determine the optimal values of parameters α and β. Next, based on the selected weight for the reconstruction loss, we fine-tune the learning rate to guide the model to iterate toward the global optimum. Finally, based on the previously determined parameters, we select suitable values for the sparse constraint parameters γ and δ, as well as the variation constraint parameter λ, to achieve optimal performance. For the Jasper Ridge dataset, we plotted curves of unmixing results under different parameter settings, as shown in Fig. 7. The detailed settings for corresponding datasets are shown in Table I. We employ the Adam optimizer with a learning rate decay of 0.9 every 20 epochs, and the number of epochs is set to 200. The results are the average of ten runs.

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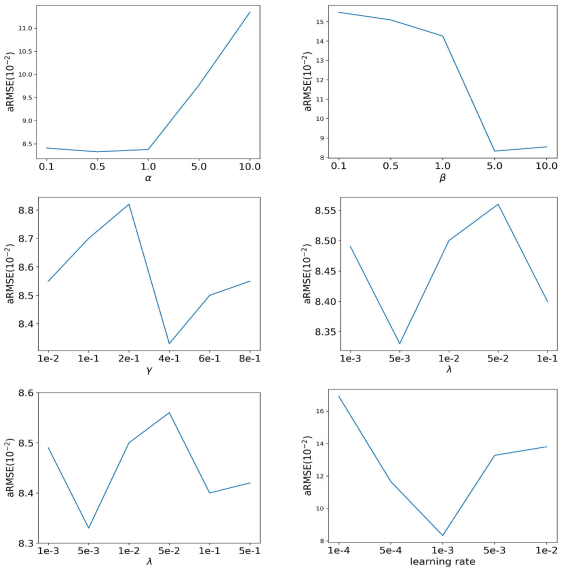


Fig. 7. RMSE as functions of the parameters α, β, γ, δ, λ, and learning rate.

TABLE I
HYPERPARAMETER SETTINGS FOR DATASETS

Parameter	Reconstruction		Sparsity		Variational	LR
	α	β	γ	δ	λ	
Synthetic	5.0	4e-1	2e-1	1e-1	5e-3	2e-3
Samson	5.0	1e-2	4e-1	1e-1	5e-3	2e-3
Jasper Ridge	5e-1	5.0	4e-1	2e-1	5e-3	1e-3
Urban	1.0	1.0	4e-1	2e-1	1e-2	1e-3
Cuprite	5e-1	5.0	4e-1	2e-1	5e-3	1e-3

D. Experiments on Synthetic and Real Datasets

1) Synthetic Dataset: In order to validate the performance of the algorithm and its sensitivity to noise, we conduct experiments in synthetic datasets with different SNRs in the range of 10, 20, and 30 dB. We present the results in the form of bar graphs and the experimental results are shown in Fig. 8. At high SNR, ALMM demonstrates its higher accuracy, which is attributed to its modeling of SV from both scaling and perturbation terms. However, its performance is greatly affected by SNR, deteriorating sharply as noise becomes severe. Considering the performance of comparison algorithms under different SNR conditions, UST-net, ULA-Net, and the proposed algorithm exhibit the best performance. Nevertheless, due to the absence of a specific noise handling module, UST-Net shows inferior endmember estimation performance at high noise levels. Furthermore, the proposed algorithm introduces global spatial information and integrates various SV factors compared to ULA-Net, hence demonstrating relatively better performance.

2) Samson Dataset: Quantitative results of abundance and endmember on the Samson dataset are shown in Tables II and III. The proposed algorithm achieves optimal results in both abundance and endmember estimation. Comparison algorithms can be categorized into those focusing on modeling SV and those incorporating spatial self-attention

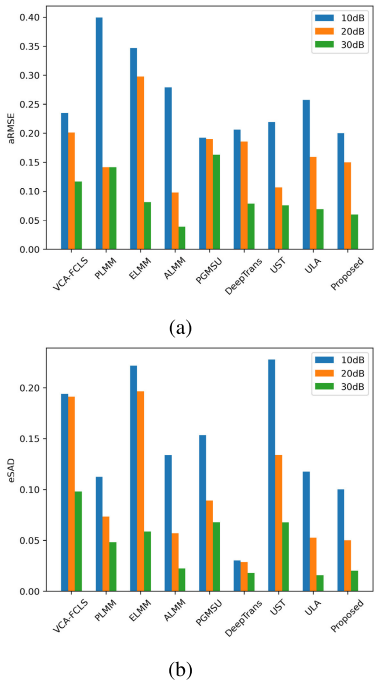


Fig. 8. Experimental results on synthetic dataset with different SNRs. (a) aRMSE. (b) eSAD.

mechanisms. Compared to the traditional VCA-FCLS algorithm, all categories show performance improvements. Moreover, according to quantitative results, SV modeling algorithms tend to suppress the influence of SV, thus enhancing abundance estimation accuracy. ALMM, ULA-Net, and the proposed algorithm all achieve outstanding results, demonstrating that introducing SV modeling can effectively mitigate its impact on abundance estimation. Additionally, models incorporating spatial self-attention mechanisms achieve better endmember extraction accuracy while maintaining high abundance precision. The proposed network designs an effective SV feature extraction strategy, leading to improved unmixing accuracy.

To further demonstrate the performance of the proposed algorithm, we draw abundance maps and endmember curves in Figs. 9 and 10. The estimated abundance maps maintain a high purity level and closely resemble the ground truth. Furthermore, the endmembers extracted by the network exhibit the highest similarity to the true endmember curves, demonstrating its outstanding performance.

3) Jasper Ridge Dataset: The quantitative results of abundance and endmember on the Jasper Ridge dataset are shown in Tables IV and V, where the proposed algorithm achieves optimal results. Especially for complexly distributed materials like tree and dirt, our work is able to estimate their abundances more accurately. Compared to the local-attention mechanism in ULA-Net, the global-attention mechanism in UST-Net and the proposed algorithm are better at capturing useful information in complex data. Furthermore, the introduction of the feedback mechanism and SV modeling further enhances unmixing accuracy. Besides, the abundance maps and endmember curves are present in Figs. 11 and 12. Clearly, our work provides more accurate estimations of the abundance of

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TABLE II
ABUNDANCE RMSE RESULTS ON SAMSON DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
Soil	0.2250	0.1206	0.0997	0.1009	0.2064	0.1090	0.1146	0.0934	**0.0628**
Tree	0.1813	0.0600	0.0878	0.0548	0.1825	0.0869	0.0906	**0.0537**	0.0588
Water	0.3388	0.0962	0.1361	0.1064	0.2838	0.1503	**0.0472**	0.1039	0.0582
Mean RMSE	0.2484	0.0923	0.1079	0.0874	0.2242	0.1154	0.0841	0.0837	**0.0599**

TABLE III
ENDMEMBER SAD RESULTS ON SAMSON DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods		VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
SAD	Soil	0.0365	0.0341	0.0267	0.0365	0.0307	0.0196	**0.0136**	0.0207	0.0208
	Tree	0.0309	0.1140	0.0954	0.0309	0.0560	0.0436	0.0691	**0.0289**	0.0535
	Water	0.1245	0.2168	0.1472	0.1245	0.2096	0.1096	0.0624	0.1206	**0.0471**
Mean SAD		0.0640	0.1216	0.0898	0.0640	0.0987	0.0576	0.0483	0.0567	**0.0383**
Mean SID		0.0197	0.1262	0.0423	0.0244	0.0644	0.0295	0.0146	0.0274	**0.0065**

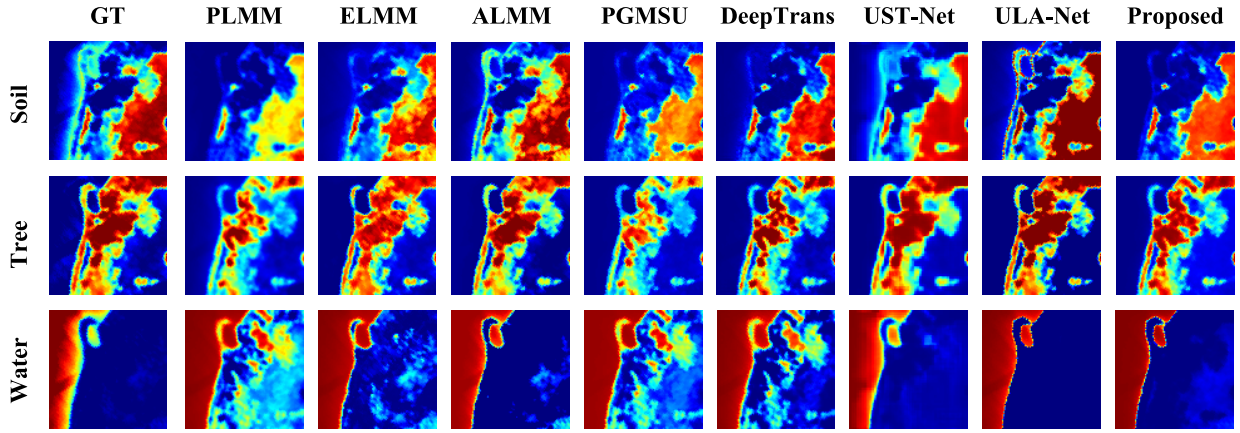


Fig. 9. Abundance map of Samson dataset.

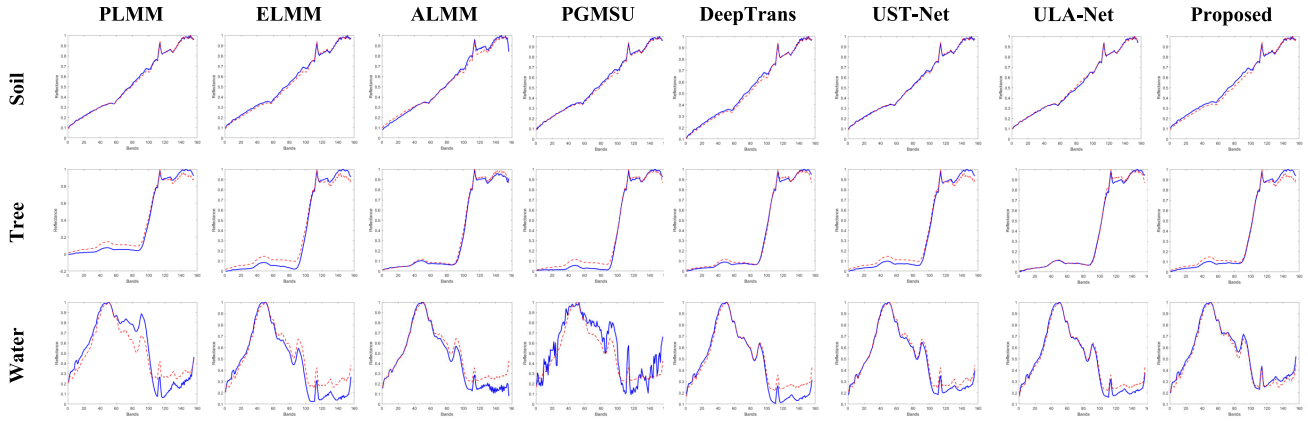


Fig. 10. Endmember curve of Samson dataset, where the red line is the true curve, and the blue line is the extraction results of different methods.

trees and soil. Additionally, we effectively distinguish between soil and roads, avoiding confusion between them. As for endmember curves, our work also presents better visualization results.

4) Urban Dataset: To provide a more comprehensive demonstration of the model's performance, we conduct experiments on the Urban dataset with six endmembers. The quantitative results are presented in Tables VI and VII, in which the proposed model achieves the best results. Based on the experimental results, ELMM, UST-Net, ULA-Net, and the proposed algorithm achieve the best performance. The ELMM introduces a scaling decoder for per-endmember scaling, demonstrating excellent performance in both abundance and endmember estimation. UST-Net extracts and fuses

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TABLE IV
ABUNDANCE RMSE RESULTS ON JASPER RIDGE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
Tree	0.1360	0.0746	0.0782	0.1247	0.1349	0.0889	0.0754	0.1065	**0.0514**
Water	0.0767	0.2740	0.1402	0.1401	0.1090	0.0911	**0.0322**	0.1324	0.0348
Dirt	0.2202	0.3386	0.1998	0.1586	0.2199	0.2843	0.1452	0.1192	**0.1030**
Road	0.2137	**0.1085**	0.1693	0.1989	0.2315	0.2937	0.1999	0.1860	0.1154
Mean RMSE	0.1617	0.1989	0.1469	0.1556	0.1738	0.1895	0.1132	0.1360	**0.0833**

TABLE V
ENDMEMBER SAD RESULTS ON JASPER RIDGE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods		VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
SAD	Tree	0.2089	0.2972	0.1376	0.2089	0.1230	0.0453	0.0838	0.1381	**0.0323**
	Water	0.1686	0.4742	0.1952	0.1686	0.1204	0.0582	**0.0305**	0.0504	0.0881
	Dirt	0.2017	0.0909	0.1312	0.2017	0.1265	0.1318	0.2594	0.1631	**0.0479**
	Road	0.4266	0.4627	0.2658	0.4266	0.4520	0.3825	**0.1100**	0.3936	0.1283
Mean SAD		0.2515	0.3313	0.1825	0.2515	0.2055	0.1545	0.1290	0.1863	**0.0741**
Mean SID		0.1435	3.6701	0.0752	0.0768	0.0679	0.0768	0.1638	0.0548	**0.0305**

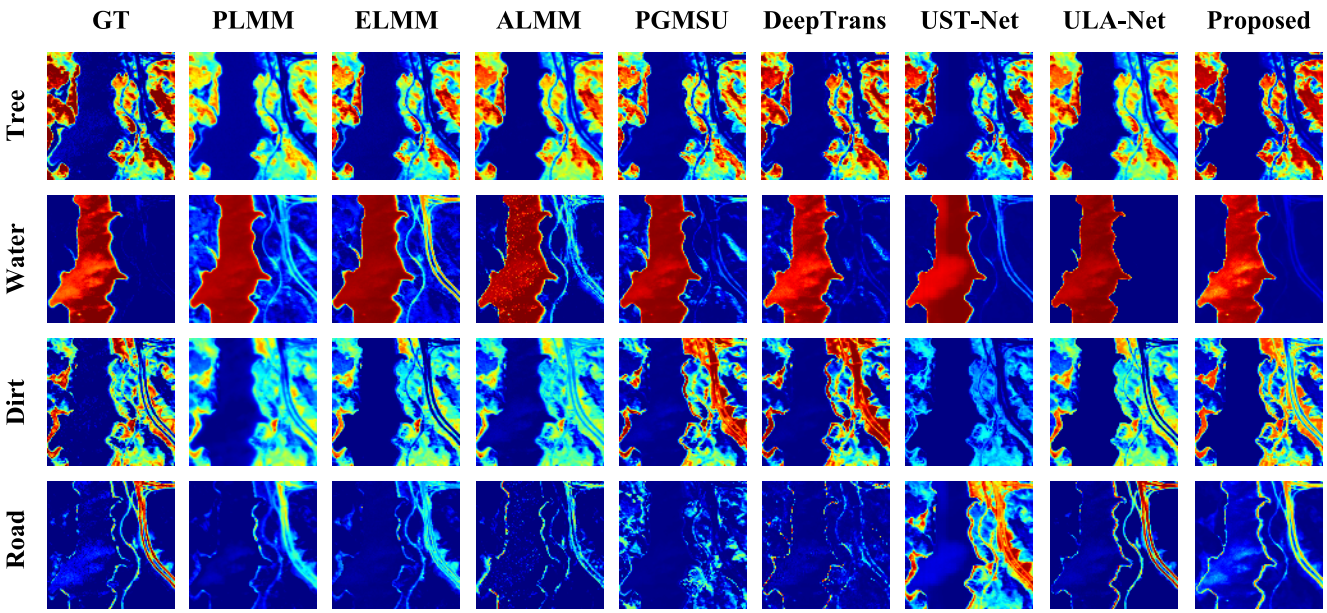


Fig. 11. Abundance map of Jasper Ridge dataset.

TABLE VI
ABUNDANCE RMSE RESULTS ON URBAN DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
Asphalt	0.1999	0.2145	0.1652	0.1730	0.1429	0.2323	**0.1206**	0.1653	0.1642
Grass	0.2037	0.2367	0.0972	0.1880	0.3212	0.3143	0.1538	0.0897	**0.0786**
Tree	0.2438	0.1694	0.0859	0.2679	0.4512	0.3134	0.0734	0.0716	**0.0584**
Roof	0.2374	0.0754	**0.0307**	0.1101	0.0814	0.1551	0.0423	0.0321	0.0366
Metal	0.0404	0.1785	0.0554	0.0601	0.0368	0.1553	0.1381	0.0616	**0.0389**
Dirt	0.1617	0.1690	0.1666	0.1316	**0.1170**	0.2236	0.1199	0.1619	0.1690
Mean RMSE	0.1747	0.1739	0.1002	0.1551	0.1918	0.2323	0.1080	0.0970	**0.0910**

features from a multiscale perspective but only considers linear mixing, resulting in superior abundance estimation but suboptimal endmember extraction due to the existence of SV. ULA-Net and the proposed algorithm are based on Transformer for feature extraction and decoders considering SV, yielding outstanding results.

The visualization results are presented in Figs. 13 and 14. Regarding the abundance maps, the proposed algorithm

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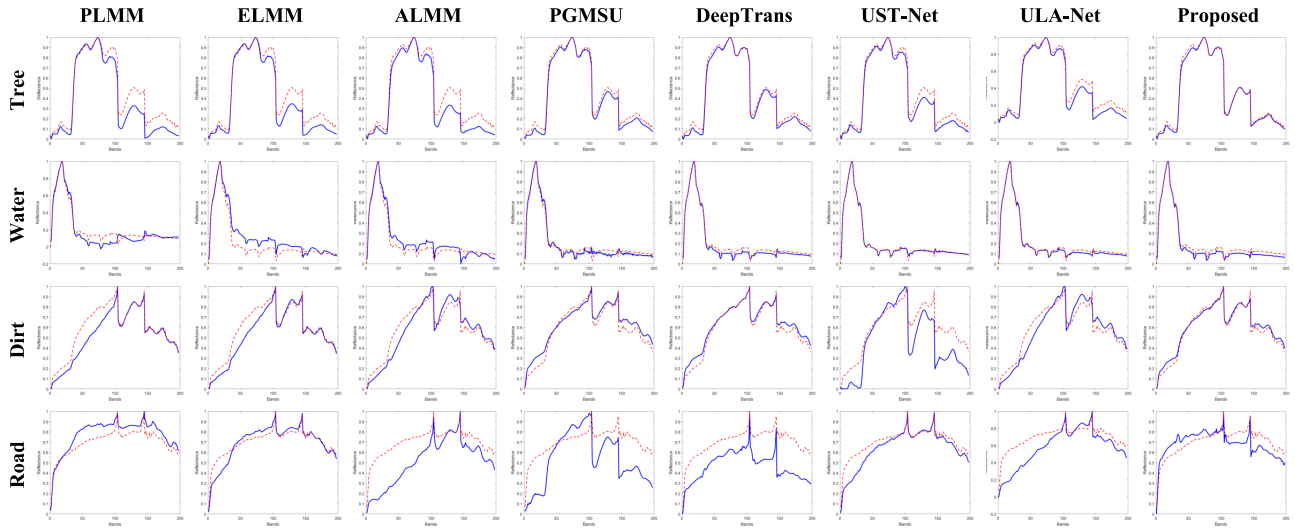


Fig. 12. Endmember curve of Jasper Ridge dataset, where the red line is the true curve and the blue line is the extraction results of different methods.

TABLE VII
ENDMEMBER SAD AND SID RESULTS ON URBAN DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods		VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
SAD	Asphalt	0.1334	0.0908	0.0846	0.1948	0.0890	0.1375	**0.0510**	0.0851	0.0851
	Grass	0.5341	0.5720	**0.0394**	0.3169	0.3088	0.2142	0.0678	0.0705	0.0711
	Tree	0.8115	0.3370	0.1329	0.6960	0.6125	0.1207	**0.1061**	0.1520	0.1246
	Roof	0.2252	0.2958	0.0996	0.2252	0.3093	0.1864	**0.0380**	0.0647	0.0700
	Metal	0.3543	0.3365	0.2210	0.3543	0.1331	0.3897	0.3172	0.1892	**0.1799**
	Dirt	0.5993	1.5070	**0.1198**	0.3461	0.1596	0.8158	0.5084	0.1801	0.1569
Mean SAD		0.4430	0.4374	0.1162	0.3556	0.2687	0.3107	0.1814	0.1236	**0.1146**
Mean SID		0.3348	1.6286	0.0258	0.2201	0.1483	0.6324	0.0862	0.0298	**0.0229**

TABLE VIII
ENDMEMBER SAD AND SID RESULTS ON CUPRITE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods		VCA-FCLS	PLMM	ELMM	ALMM	PGMSU	DeepTrans	UST-Net	ULA-Net	Proposed
SAD	Alunite	0.1009	0.1105	0.1659	0.1238	0.2035	0.1224	0.1157	**0.0879**	0.1111
	Andradite	0.0943	0.0628	0.1155	0.0731	0.0932	0.0742	0.0795	0.0821	**0.0616**
	Buddingtonite	0.1075	0.0887	0.1439	0.1130	0.1037	0.1179	0.1179	0.0784	**0.0726**
	Dumortierite	0.1202	0.0889	0.1103	0.1111	0.1279	0.1121	0.1047	0.0983	**0.0741**
	Kaolinite1	0.0921	0.0827	0.0801	0.0784	0.1575	0.1096	0.0833	**0.0755**	0.0792
	Kaolinite2	0.0939	0.0815	0.0905	0.0753	0.0647	0.1049	0.1031	0.0903	**0.0606**
	Muscovite	0.0985	0.1429	0.1396	0.0942	0.1248	0.1908	0.2331	0.0940	**0.0792**
	Montmorillonite	0.1461	0.0945	0.1448	0.0918	0.0544	0.1212	0.1239	0.0685	**0.0548**
	Nontronite	0.2068	0.1172	0.1872	0.0968	0.1071	0.1423	0.1568	0.0942	**0.0690**
	Pyrope	0.2973	0.0790	0.1181	0.1007	0.1551	0.0571	0.0779	0.0688	**0.0562**
	Sphene	0.3497	0.0626	0.3307	0.2237	0.2106	0.3135	0.0659	0.0768	**0.0609**
	Chalcedony	0.1078	0.2662	0.1715	0.1960	0.1201	0.5478	0.2765	0.2498	**0.0867**
Mean SAD		0.1512	0.1065	0.1499	0.1148	0.1269	0.1678	0.1282	0.0970	**0.0722**
Mean SID		0.0386	0.0407	0.0362	0.0316	0.0221	0.0782	0.0685	0.0194	**0.0086**

achieves the best results for grass, tree, and roof. However, in the Urban dataset, estimating and extracting the abundance and endmembers of metal and dirt pose challenges. Although UST-Net provides a reasonable estimation of the metal abundance map, it confuses metal with grass and trees, resulting in significant misclassifications on the right side of the abundance map. In contrast, the proposed algorithm maintains high purity and avoids misclassification of endmembers due to the sparsity constraint. This results in an overall lower error but at the cost of some loss in detail. The endmember extraction results are consistent with the above, as the proposed algorithm aims to extract representative spectra while minimizing confusion between different endmembers.

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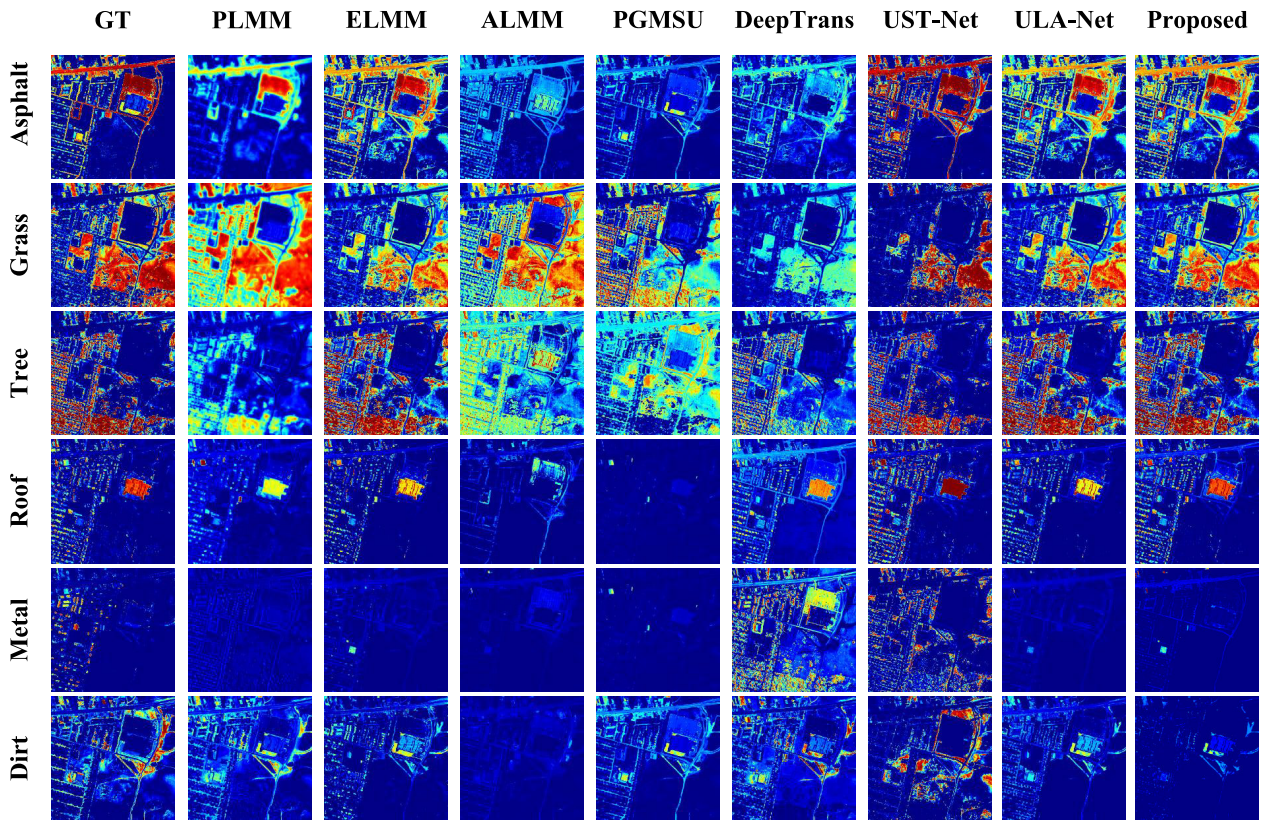


Fig. 13. Abundance map of Urban dataset.

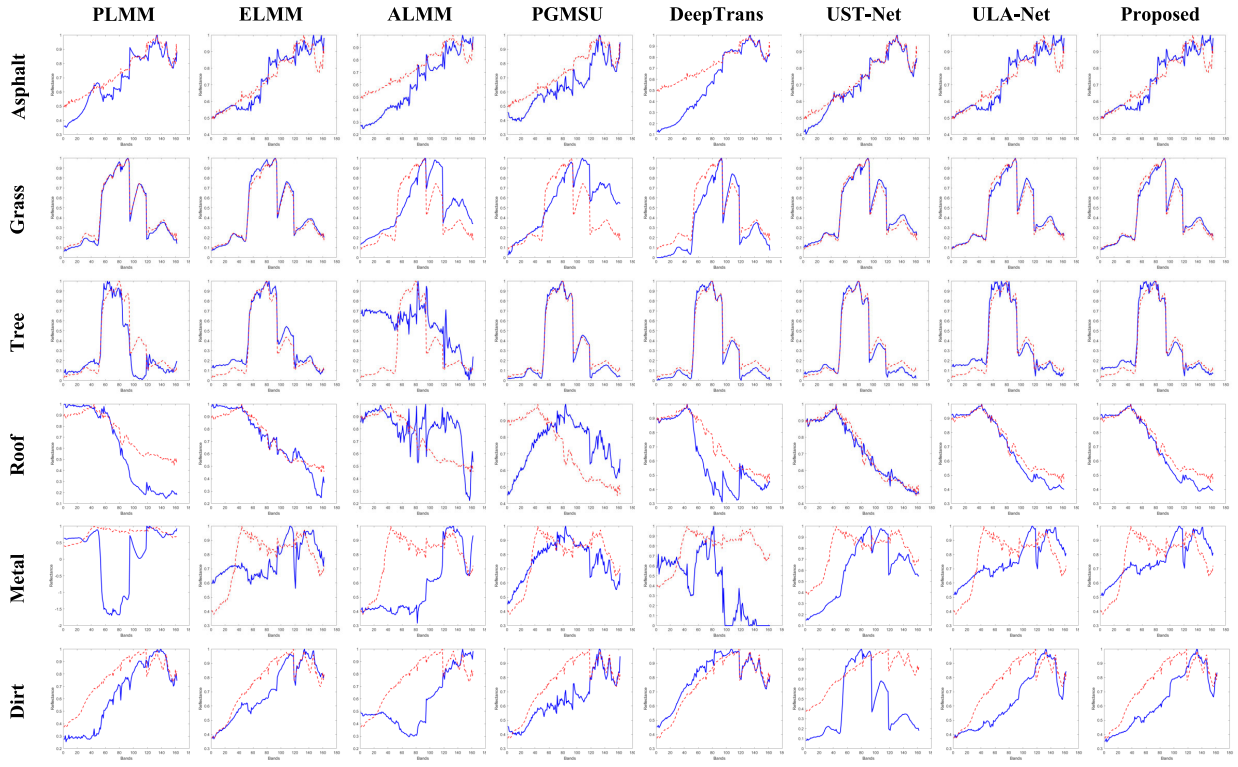


Fig. 14. Endmember curve of Urban dataset, where the red line is the true curve and the blue line is the extraction results of different methods.

5) Cuprite Dataset: To validate the performance of the model in extracting endmembers, we conduct experiments on the Cuprite dataset. The imaging area of this dataset primarily consists of mining regions, with 12 endmembers representing

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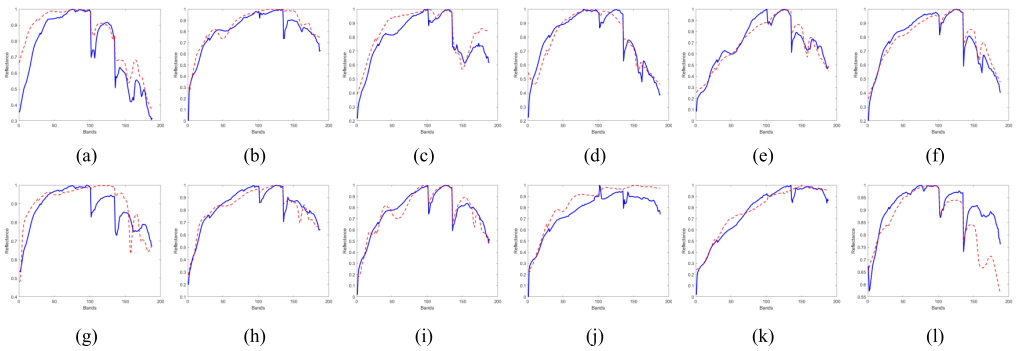


Fig. 15. Endmember curve of Urban dataset extracted by the proposed FSVA-Net, where the red line is the true curve and the blue line is the extraction results of the proposed methods. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite1. (f) Kaolinite2. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

various minerals that exhibit high similarity. The endmember extraction results are shown in Table VIII and Fig. 15, and we measure the distance between the extracted endmembers and the ground truth using both SAD and spectral information divergence. The proposed algorithm achieves the best results in terms of quantitative metrics. From the visualization results, the proposed algorithm is capable of extracting discriminative spectra that are similar to the ground truth.

E. Ablation Analysis

The proposed model consists primarily of three key components: a framework based on the feedback mechanism, an SVA module, and a generative ALMM-based decoder. The framework based on feedback mechanisms grants the model with the ability to capture SV information from linearly reconstructed residuals. The SVA module is guided by the features that emphasize SV information after feedback and can effectively extract valid information based on correlation from a global perspective. The generative ALMM-based decoder models SV from both scaling and perturbation perspectives and introduces a probability generative mechanism, enhancing the model's ability to model complex distributions.

To explore the roles of each module and validate their effectiveness, we conducted ablation experiments. Regarding the feedback framework, we compared the performance with and without the feedback branch. For the SVA mechanism, as it serves as a feature extraction module, we substituted it with a Swin Transformer block. For the decoder, we compared the influence of the probability generative mechanism on unmixing performance. The results of the ablation experiments are shown in Table IX.

Analyzing the results in Table IX, we can demonstrate the efficacy of each module. When we ablate the proposed modules (Case 1), the main branch relies on the Swin Transformer and linear layers for unmixing. Comparing Cases 1 and 2, we observe that incorporating SV modeling in the decoder leads to decreased unmixing accuracy. This demonstrates the need for an effective feature extraction module in the design of ALMM-based decoders. Without such a module, there is a risk of making inaccurate estimations when it comes to capturing SV. As for Cases 2 and 3, we evaluate whether the residual feedback mechanism effectively guides the modeling of SV, and the results confirm its effectiveness. To validate the ability to explore SV-related features using the SVA module, we conducted experiments with Cases 3 and 5. The results demonstrate that the SVA module can effectively utilize the feedback-enhanced feature and achieve better unmixing performance. Furthermore, we compare the performance of estimating perturbation terms in the data using linear layers versus probabilistic generative models. It is evident that the probabilistic generative model can fit more complex distributions and adapt to the effects caused by SV factors in real-world scenarios.

TABLE IX
QUANTITATIVE RESULTS ON ABLATION STUDY. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Case	Feedback	SVA	G-ALMM	aRMSE	eSAD
1	×	×	×	0.1262	0.0889
2	×	×	✓	0.1464	0.0904
3	✓	×	✓	0.1116	0.0873
4	✓	✓	×	0.0964	0.0822
5	✓	✓	✓	**0.0833**	**0.0741**

V. CONCLUSION

In this article, we propose an FSVA-Net, where we design a feedback branch to model the SV based on ALMM. One of the distinguishing features of the proposed network is that it makes more contributions to separating SV information from the pure data, leveraging a feedback mechanism to augment SV information, and mining useful information from feature maps based on the SVA module. Additionally, to address the limited modeling capacity of the variable dictionary in ALMM, we introduce a generative model, which can generate perturbations that better match real-world scenarios. Experimental results confirm the effectiveness of the proposed algorithm and the efficacy of introducing the feedback branch. Future work will consider incorporating orthogonal subspace projection to explore SV-related information and design a more efficient model of SV. Additionally, to enhance the model's ability to reconstruct details in the abundance maps, we will consider designing a loss function specifically aimed at improving the recovery of detailed information.

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